9. **(a)** Define *magnetic flux density*.

.....

.....

[2]

.....

(b) Fig. 9 below shows a current carried by a square coil GHJK of 500 turns and side 5.0 cm. The coil is suspended vertically in a uniform horizontal magnetic field of flux density $B = 2.0 \times 10^{-4}$ T. The plane of the coil is at angle $\theta = 35°$ to B.

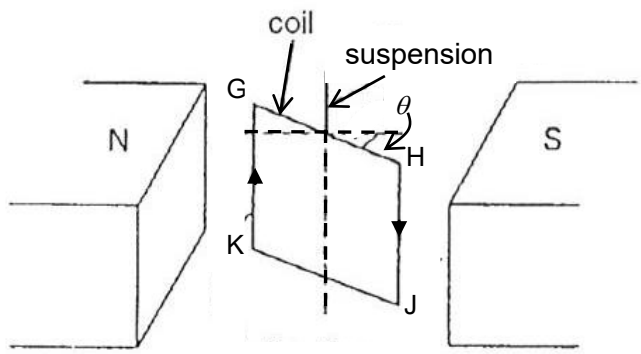

Fig. 9

(i) Indicate, in Fig. 9, the direction of the magnetic force acting on the side JK of the coil. [1]

(ii) The current in coil GHJK is 0.40 A. Calculate the magnetic force acting on side JK of the coil.

magnetic force = ... N [2]

(iii) Calculate the torque acting on the coil GHJK.

torque = ... N m [3]

(c) State the *laws of electromagnetic induction*.

.....

.....

.....

.....

..... [2]

(d) The coil GHJK in Fig. 9 is disconnected from its e.m.f. source. The plane of the coil is at angle $\theta = 35°$ to B.

(i) Calculate the magnetic flux linkage through the coil.

magnetic flux linkage = ... Wb [2]

(ii) The coil is then rotated at 100 Hz about the vertical suspension.

1. Calculate the magnitude of the maximum induced e.m.f. in the coil.

maximum induced e.m.f. = ... V [3]

2. The coil is now acting as an AC generator. It is connected to a heater of resistance 5.0 Ω. Assuming the coil has negligible resistance, calculate the power generated in the heater.

power = ... W [3]

3. If the heater in **(d)(ii)2** is also connected in series with an ideal diode, determine the power generated in the heater.

power = ... W [2]

End of Paper